Probability:

Probability is the measure of the likelihood of an event occurring,

There are total two events

1) **independent events** and **dependent events** are crucial concepts that describe how the occurrence of one event affects the probability of another event. Here's a detailed explanation of each:

Independent Events

Two events are considered **independent** if the occurrence or non-occurrence of one event does not affect the probability of the other event occurring. In other words, the events are independent if knowing the outcome of one event does not change the probability of the other event.

1) Toss Coin
2) Rolling Dice

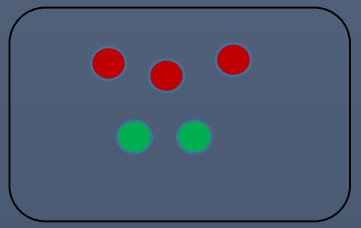
Rolling A dice

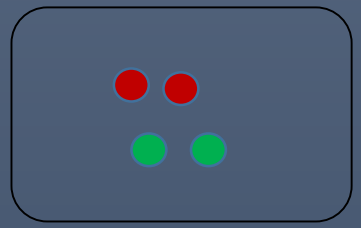


$$\Pr(1) = \frac{1}{6} \qquad \Pr(2) = \frac{2}{6} \qquad \Pr(3) = \frac{3}{6}$$

Dependent Events:

Two events are considered **dependent** if the occurrence or non-occurrence of one event affects the probability of the other event occurring. In other words, the events are dependent if the probability of one event is influenced by the outcome of the other event.





$Pr(R)\frac{3}{5}$ $Pr(R)\frac{2}{4}$

$Pr(G)\frac{2}{5}$ $Pr(G)\frac{2}{4}$

$$\Pr(\text{R \& G}) = p(R) * p\left(\frac{G}{R}\right) \qquad \Pr(\text{G \& R}) = p(G) * p\left(\frac{R}{G}\right)$$

If R = A & G = B

$$\Pr(\text{A \& B}) = p(A) * p\left(\frac{B}{A}\right) \qquad \Pr(\text{B \& A}) = p(B) * p\left(\frac{A}{B}\right)$$

$$p(A) * p\left(\frac{B}{A}\right) = p(B) * p\left(\frac{A}{B}\right)$$

$$p\left(\frac{B}{A}\right) = \frac{p(B) * p\left(\frac{A}{B}\right)}{P(A)}$$

	X1 Covid	X1 Cold	Yes Fever
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

$$p\left(\frac{B}{A}\right) = \frac{p(B) * p\left(\frac{A}{B}\right)}{P(A)}$$

Create Equation for Yes
Hence replace B with y and A WITH $x_1x_2 \dots x_n$

$$\mathrm{p}\frac{y}{x_1x_2 \dots x_n} = \frac{p(y) * p(\frac{x_1x_2 \dots x_n}{y})}{p(x_1x_2 \dots x_n)}$$

Create Equation for No
Hence replace B with n and A WITH $x_1x_2 \dots x_n$

$$\mathrm{p}\frac{n}{x_1x_2 \dots x_n} = \frac{p(n) * p(\frac{x_1x_2 \dots x_n}{n})}{p(x_1x_2 \dots x_n)}$$

$$\mathrm{p}\frac{y}{x_1x_2 \dots x_n} = \frac{p(y) * p\left(\frac{x_1}{y}\right) * p\left(\frac{x_2}{y}\right) * p\left(\frac{x_n}{y}\right)}{p(x_1)p(x_2) \dots p(x_n)}$$

$$\mathrm{p}\frac{n}{x_1x_2 \dots x_n} = \frac{p(n) * p\left(\frac{x_1}{n}\right) * p\left(\frac{x_2}{n}\right) * p\left(\frac{x_n}{n}\right)}{p(x_1)p(x_2) \dots p(x_n)}$$

$$\mathrm{p}\frac{y}{x_1x_2 \dots x_n} = p(y) * p\left(\frac{x_1}{y}\right) * p\left(\frac{x_2}{y}\right) * p\left(\frac{x_n}{y}\right)$$

$$\mathrm{p}\frac{n}{x_1x_2 \dots x_n} = p(n) * p\left(\frac{x_1}{n}\right) * p\left(\frac{x_2}{n}\right) * p\left(\frac{x_n}{n}\right)$$

	X1 Covid	X2 Cold	Yes Fever
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = p(y) * p\left(\frac{x_1}{y}\right) * p\left(\frac{x_2}{y}\right) * p\left(\frac{x_n}{y}\right) \qquad \mathrm{p}\frac{n}{x_1x_2\dots x_n} = p(n) * p\left(\frac{x_1}{n}\right) * p\left(\frac{x_2}{n}\right) * p\left(\frac{x_n}{n}\right)$$

$$Fever(Yes) = \frac{7}{10} \quad Fever(No) = \frac{3}{10}$$

$$Covid(Yes) = \frac{4}{7} \quad Covid(No) = \frac{2}{3}$$

$$Cold(Yes) = \frac{3}{7} \quad Cold(No) = \frac{2}{3}$$

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = (\frac{7}{10}) * \left(\frac{4}{7}\right) * \left(\frac{3}{7}\right)$$

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = (\frac{84}{490})$$

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = 0.171$$

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = (\frac{3}{10}) * \left(\frac{2}{3}\right) * \left(\frac{2}{3}\right)$$

$$\mathrm{p}\frac{n}{x_1x_2\dots x_n} = (\frac{12}{90})$$

$$\mathrm{p}\frac{y}{x_1x_2\dots x_n} = 0.133$$